



Assignment 2



Instructions

Enterprise Architecture Design Assignment 2

Marks: 25%

Due date: upload to Brightspace by Sunday 24th May @ 23.59

Word limit: Project report max 1500 words (excluding diagrams/appendices)

Submission format: Brightspace **upload (zipped folder)** + PDF report + screencast (link to student OneDrive shared video)

Note: *This is an individual assignment, all work must be your own. The University Plagiarism rules apply.*

Overview

You are continuing work on a cloud-native e-commerce platform deployed on Kubernetes. The system is now functionally complete, but before it can be considered suitable for a more realistic enterprise setting, it must be reviewed and improved from a security and operations perspective.

Your task is to review, improve, test, and demonstrate the platform in three areas:

- secure design
- logging, monitoring, and observability
- controlled security testing of the container and Kubernetes environment

You will be assessed on the quality of your evidence, judgement, and reasoning, not simply on whether tools were run or fixes were applied. This follows the same approach as Assignment 1.

Requirements

1. Secure design review

Review the current solution and identify meaningful security risks or weaknesses.

Your review should consider areas such as:

- exposed entry points
- internal service-to-service trust assumptions

- secrets and credential handling
- container and workload security
- network exposure
- dependency and image risk

You must propose realistic improvements or mitigations.

2. Architecture diagrams and design communication

You must include two architecture diagrams as part of your submission.

Diagram A: Current-state architecture and trust boundaries

Show the deployed system as it currently exists, including:

- entry points
- main services
- database or persistence layer, where used
- internal communication paths
- trust boundaries or other security-relevant boundaries

Diagram B: Improved-state architecture

Show the revised design after your security and observability improvements.

This should include relevant additions such as:

- improved logging and monitoring flows
- security controls
- testing or scanning boundaries
- any changes to network exposure or trust assumptions

3. Logging, monitoring, and observability

Improve the platform's operational visibility.

The system must produce logs that are useful for investigation. You must also add monitoring or metrics that help explain system behaviour.

You must demonstrate at least one realistic scenario in which the added observability helps diagnose or explain a problem.

4. Security testing

Perform controlled security testing on your own deployed lab environment.

Your testing must include:

- vulnerability or image scanning
- configuration or posture review
- at least one deeper test of platform or container exposure

You must use and interpret some of the tools listed in the referenced Docker/container security tools article, such as image scanners or posture-review tools.

You must explain what the findings mean and identify what should be fixed first.

5. Findings and remediation

Summarise the key findings from your review and testing.

You must:

- prioritise the issues identified
- explain what you changed
- explain what you would change next
- justify why those actions should be prioritised

Deliverables and marking scheme (100 marks)

Part 1 (25%): Secure design review and architecture

Include a design and security review section in the final report, supported by both required architecture diagrams.

Assessment will consider:

- clarity of the review of current risks and weaknesses
- accuracy of the architecture diagrams in relation to the implemented system
- realism of the proposed mitigations and design improvements

Part 2 (25%): Observability implementation

Submit repository updates, manifests/configuration, and supporting evidence.

Assessment will consider:

- whether useful logging and monitoring were added
- whether at least one realistic scenario was demonstrated and clearly explained

Part 3 (30%): Security testing

Submit a testing report including captured outputs (screenshots or copied command outputs) and any scripts or commands used.

Assessment will consider:

- appropriate use of container or Kubernetes security tools

- meaningful interpretation of findings
- clear prioritisation of remediation actions

Part 4 (20%): Project report and screencast

Submit a report of no more than 1,500 words and a screencast.

Assessment will consider:

- the overall quality and clarity of the report
- the effectiveness of the screencast in demonstrating the solution, findings, and improvements

Recommended screencast length: 5 to 10 minutes, with clear voice-over.

Submissions

No submissions yet. Drag and drop to upload your assignment below.

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Activity Details

Task: Submit to complete this assignment



Due 24 May at 11:59 PM

Last Visited 15 May 2026 11:56 AM